Do geospatial foundation models detect surface change better than spectral indices? A noise-calibrated comparison on forest loss and Himalayan glacial lakes

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Abstract

Foundation models pretrained on satellite imagery are promoted for change detection but rarely compared with simple spectral indices under a common standard. We score reflectance, spectral indices, a signed NDVI-decrease baseline and two foundation-model embeddings (Prithvi-EO-2.0 and an SSL4EO-S12 ResNet-50) with one rule: the change between two periods in units of the scene-to-scene noise, which works for feature vectors of any length. We evaluate against independent products: forest loss in five Indian areas (Hansen Global Forest Change) and water change at four Nepal glacial lakes (Sentinel-1 radar). On forest loss no foundation model beats the unsupervised signed NDVI baseline in any area; Prithvi ties it in two (average precision 0.69 against 0.72 in Bastar, 0.52 against 0.54 in Kodagu) and is clearly worse in three. A trained linear readout of Prithvi beats the baseline in two areas, but a readout of six raw bands does about as well. On glacial lakes Prithvi (0.63) and raw bands (0.64) beat the MNDWI water index (0.45) and tie each other, on two changing lakes that hold 49 of 59 positive patches. In a placebo test on twelve areas the embeddings raise 0 to 3.4% false alarms against up to 41% for indices, a calibration property rather than detection skill. Code, scene lists and checksums are released.

1 Introduction

Detecting where a landscape has changed between two dates is a basic task in remote sensing, and foundation models trained on millions of Sentinel-2 scenes [1], [2] are now offered as a general-purpose answer. Evidence that they beat the long-standing alternative, a spectral index such as NDVI thresholded on its difference [3], [4], [5], is thin, usually reported on benchmarks [6], [7] rather than against independent reference data, and almost never with the noise between acquisitions taken into account. Two images of the same place never agree exactly: clouds, haze, sun angle and season change every pixel. A method that reports "change" without a notion of this noise cannot say how much of its output is real.

We ask a narrow question: when every method is judged against the same scene-to-scene noise and the same reference data, do foundation-model embeddings detect real surface change more reliably than spectral indices, and does the answer hold across fields? We contribute (i) a noise-calibrated change score for any feature vector, (ii) a pre-specified evaluation in two fields (five forest areas, four glacial lakes) with spatial block bootstrap intervals and a reference-free placebo test, and (iii) a negative-leaning result reported as such: no foundation model beat the best simple baseline when scored without labels, and a trained linear readout of the frozen features, tried afterwards, did no better than the same readout of raw reflectance.

2 Data and reference products

Optical. Sentinel-2 L2A scenes from the Element 84 Earth Search catalogue, six bands (blue, green, red, narrow NIR, SWIR1, SWIR2), cloud and snow masked with the scene classification layer. Forest pe-

riods use three clear scenes each inside one February to March window. At three lakes the earlier period had one clear scene and the later three, so their noise scale comes from the later period alone; the fourth lake has three in both.

Forest loss. Five areas of about 7.6 by 8 km, compared between February to March 2021 and 2025: Bastar and Odisha (central-east India), Kodagu and Wayanad (Western Ghats), Garo Hills (Meghalaya) and the Shivalik foothills (Uttarakhand). They were registered before any score was computed (`studies/forest_expansion_prereg.yaml`). Boxes centred on a stated point failed a label-only screen of at least 150 positive patches, so we amended the registration, still before any scene or score: each box is the one with most positive patches inside a 0.4 degree window, chosen from Hansen labels alone. A fourth region, the Nepal Terai near Chitwan, failed even with its best box (40 positives) and is not evaluated. The boxes therefore over-represent active clearing. The reference is Hansen Global Forest Change v1.12 [8]: a patch is positive if at least 10% of it was lost in 2022 to 2024 and clean if under 2% was lost in 2021 to 2024; patches between are set aside, and only patches with at least 50% forest cover are eligible.

Glacial lakes. The reference is the change in Sentinel-1 water frequency per patch (Sentinel-1 RTC from Microsoft Planetary Computer, scenes within 20 days of each window, usable orbits only). A patch is positive if at least 25% of it changed water state and clean if under 5% did. Radar is a different sensor from the optical data under test, so the reference is not circular, but it has its own errors. Three pilot lakes (Tsho Rolpa, Imja Tsho, Thulagi; October to November 2018 against 2024) held 44 positive patches, 34 at one lake. We therefore screened all 30 lakes of our Hi-MAG-based inventory [9] with the radar reference alone, before any optical scene or score, and added each other lake with at least 10 positive patches (`studies/lake_expansion_amendment.yaml`). Two passed, but neither had a clear 2018 optical scene, so the earlier period was moved to 2020 for added lakes; the repeated screen kept one (15 positives) and dropped the other (7). All four lakes were chosen or kept because the radar shows change at them.

All references are other satellite products and are treated as agreement targets, not ground truth.

3 Methods

3.1 A change score in units of noise

Each scene is reduced to one feature vector per 16×16 pixel patch (160 m). Let $\bar{f}_A$ and $\bar{f}_B$ be the mean feature vectors over the n_A and n_B scenes of the earlier and later period, and $D^2 = \|\bar{f}_B - \bar{f}_A\|^2$. The noise is estimated from pairs of scenes in the same period,

$$\sigma^2 = \mathrm{mean}_{i<j} \ \tfrac{1}{2}\|f_i - f_j\|^2,$$

pooled over a 3×3 patch neighbourhood because three scenes give only three pairs. The score is

$$z^2 = \frac{D^2}{\sigma^2\,(1/n_A + 1/n_B)},$$

about 1 when nothing has changed, whatever the length or scale of the features. The HabitatShift pipeline [10] already tests whether the change at a pixel exceeds the scene-to-scene spread of its own period; this score is that test written for feature vectors of any length.

3.2 Methods and evaluation

All methods use the same patches and the same z: single indices (NDVI, NDWI, MNDWI, NDBI), five indices together, six standardised raw bands, *Prithvi-EO-2.0-300M* [1] (encoder only, single time step, six bands, 1,024 numbers per patch) and an *SSL4EO-S12 ResNet-50* [2] using layer-3 features of the RGB bands only. Because z is unsigned, forest loss also gets a *signed NDVI-decrease* baseline, which uses the sign of the NDVI change and which an embedding cannot provide. The design was fixed before any score was computed. We report average precision (AP) and ROC-AUC against the reference labels, with 95% intervals from a spatial block bootstrap (blocks of 8×8 patches, 500 resamples) and paired differences from a named baseline resampled on the same blocks. In the placebo test one scene of a period is compared with the rest of the same period; nothing has changed, so every patch with $z > 3$ is a false alarm. This needs no reference and was run on twelve areas.

4 Results

4.1 Forest loss: no foundation model beats a signed NDVI baseline

Figure 1 and Table 1 show AP against Hansen loss. No foundation model beats the signed NDVI-decrease score in any area. Prithvi is not distinguishable from it in Bastar (0.69 against 0.72) and Kodagu (0.52 against 0.54) and clearly worse in Odisha (0.56 against 0.76), Garo Hills (0.48 against 0.55) and the Shivalik foothills (0.64 against 0.77). The RGB-only SSL4EO model is clearly worse in four areas. Six raw bands tie the baseline in three areas and are far below it in Odisha and the Shivalik foothills, where Prithvi beats raw bands because unsigned raw-band scores are near chance (AP 0.19, AUC 0.47); the leaf state of deciduous sal forest in February to March can differ between years, which we did not test. At the fixed rule $z > 3$ the embeddings catch more true loss but flag many more clean patches (Bastar: Prithvi 48% recall at 5.6% false positives, signed NDVI 24% at 0.6%); at a matched false-positive rate the baseline is at least as good.

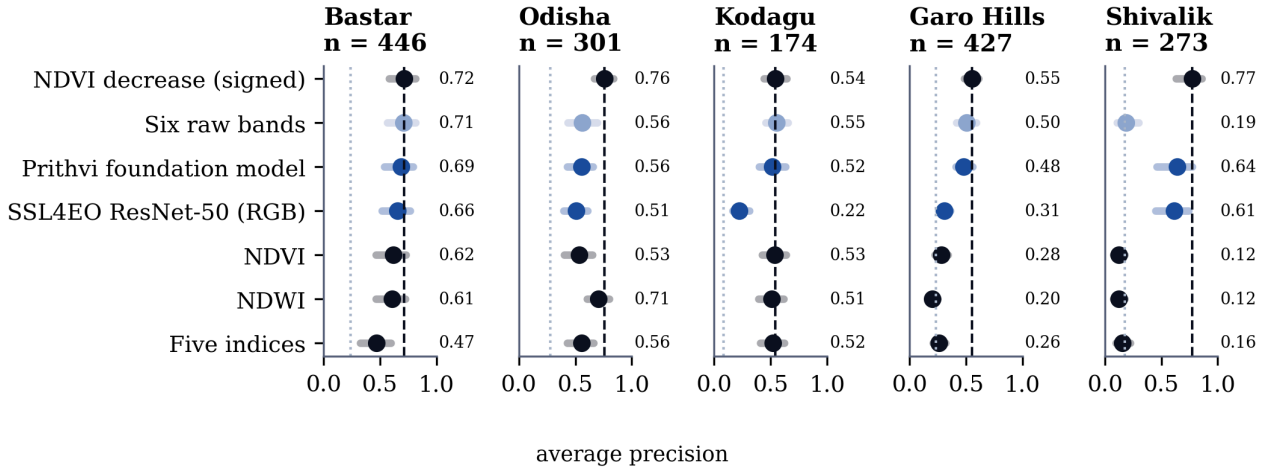


Figure 1. Average precision against Hansen forest loss, with 95% block-bootstrap intervals. Blue: foundation models; light blue: raw bands; dark: spectral indices. Dashed line: signed NDVI-decrease baseline; dotted line: chance.

Table 1. Forest loss in five areas, average precision (ROC-AUC). The signed NDVI-decrease score is the baseline; * means the paired difference from it, resampled on the same spatial blocks, has a 95% interval that excludes zero (every such difference is negative, so the method is worse). Positive patches: 446, 301, 174, 427 and 273.

Score	Bastar	Odisha	Kodagu	Garo Hills	Shivalik
NDVI decrease (signed)	0.72 (0.84)	0.76 (0.84)	0.54 (0.92)	0.55 (0.78)	0.77 (0.90)
Six raw bands	0.71 (0.85)	0.56 (0.76) *	0.55 (0.93)	0.50 (0.73)	0.19 (0.47) *
Prithvi-EO-2.0	0.69 (0.88)	0.56 (0.80) *	0.52 (0.91)	0.48 (0.78) *	0.64 (0.90) *
SSL4EO ResNet-50 (RGB)	0.66 (0.83)	0.51 (0.75) *	0.22 (0.76) *	0.31 (0.66) *	0.61 (0.89) *
NDVI (unsigned)	0.62 (0.78) *	0.53 (0.73) *	0.53 (0.92) *	0.28 (0.54) *	0.12 (0.32) *
NDWI	0.61 (0.74) *	0.71 (0.83) *	0.51 (0.90) *	0.20 (0.41) *	0.12 (0.31) *
Five indices	0.47 (0.64) *	0.56 (0.73) *	0.52 (0.91)	0.26 (0.52) *	0.16 (0.44) *

4.2 Glacial lakes: models beat the water index, and so do raw bands

Figure 2 and Table 2 give the lake result with paired intervals against MNDWI. Prithvi (AP 0.63) and six raw bands (0.64) beat MNDWI (0.45), with intervals that exclude zero (+0.06 to +0.22 and +0.06 to +0.24); the RGB-only model, NDVI and the five indices do not. Prithvi and raw bands are indistinguishable, so the gain over MNDWI is not evidence that pretraining helps. Noise calibration helps raw bands (0.64 with z, 0.20 with the raw distance) and Prithvi (0.63 against 0.35) but not the RGB-only model (0.43 against 0.70; z was the pre-specified score). The result rests on 59 positive patches: 34 at Imja Tsho, which grew from about 1.1 to 1.7 km^2, 15 at the added lake and 10 at the two lakes that stayed about the same size. It is a test on two changing lakes, the intervals are wide (Prithvi AP 0.63, 95% interval 0.22 to 0.85), and lakes were not examined separately.

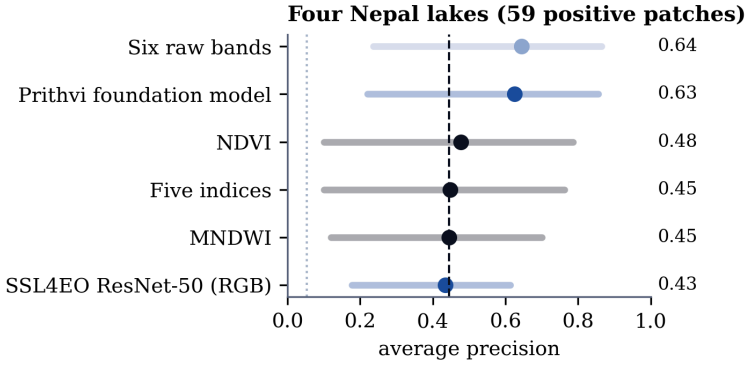


Figure 2. Average precision against the radar water-change reference at four Nepal lakes.

Table 2. Glacial lakes, average precision (ROC-AUC) and paired difference from MNDWI (95% interval; * excludes zero). 59 positive and 1,037 clean patches.

Score	AP (AUC)	Paired diff. from MNDWI
Six raw bands	0.64 (0.90)	+0.06 to +0.24 *
Prithvi-EO-2.0	0.63 (0.88)	+0.06 to +0.22 *
NDVI	0.48 (0.73)	-0.07 to +0.11
MNDWI	0.45 (0.76)	baseline
Five indices	0.45 (0.75)	-0.07 to +0.10
SSL4EO ResNet-50 (RGB)	0.43 (0.87)	-0.23 to +0.23

4.3 Placebo test: embeddings are quiet, indices are not

In the placebo test on twelve areas (Figure 3) the two embedding models raise 0 to 3.4% false alarms at $z > 3$ (medians under 0.5%), whereas indices and raw bands raise up to 41% (medians 3 to 13%). This is not better detection: on Bastar, NDVI flags 23% of patches in the placebo test but only 1% of Hansen's clean patches, so the placebo rate depends on how alike the noise is in the two periods. A likely reading is that embeddings vary little between scenes taken days apart but respond to year-to-year differences (season, illumination, phenology) that the references do not count as change; we did not test this.

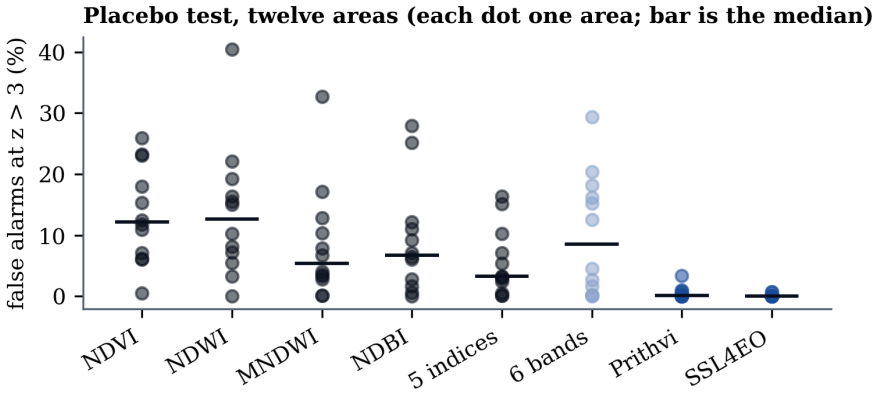


Figure 3. False alarms at $z > 3$ when one scene is compared with the rest of its own period, twelve areas.

4.4 A trained readout of the frozen features (exploratory)

A reviewer can fairly ask whether embeddings do better when read out with labels. After the main evaluation we trained an L2-regularised logistic regression on the signed noise-calibrated change vector (the change in the feature vector over its noise scale, so its length is z), leaving one forest area out in turn and choosing the regularisation only inside the four training areas (`studies/probe_design.yaml`, written before the run). A probe on six raw bands is the control. The paired comparison with the six-band probe and a wider regularisation grid were added after the first run showed they were needed, and are exploratory. Table 3 gives the result.

The Prithvi probe beats the unsupervised signed NDVI baseline in Bastar and the Shivalik foothills and ties it elsewhere, so a trained readout recovers an advantage that unsupervised z lacks. But the six-band probe does about as well: it is better than the Prithvi probe in Kodagu and, narrowly, Bastar, and in no area is the Prithvi probe clearly better under both grids. The RGB-only probe is worse than the baseline in four areas. The chosen regularisation sat at the lowest grid value in every fold; with a wider grid the Prithvi probe beats the baseline in the same two areas and, only just, in Odisha, and the comparison with the six-band probe stays unclear. The gain comes from labels and the sign of the change, not from pretraining. A pooled leave-one-lake-out probe on the four lakes was not robust (Prithvi probe AP 0.77 against 0.61 for six bands on the original grid, 0.54 against 0.39 on the wider grid, intervals spanning zero in one case) and is not read as evidence.

Table 3. Linear probe on forest loss, average precision of the held-out area (difference with 95% block-bootstrap interval; * excludes zero).

Area	Signed NDVI (no labels)	Prithvi probe	Prithvi probe minus signed NDVI	Six-band probe	Prithvi probe minus six-band probe
Bastar	0.72	0.77	+0.02 to +0.11 *	0.81	-0.07 to -0.00 *
Odisha	0.76	0.79	-0.01 to +0.07	0.76	-0.00 to +0.08
Kodagu	0.54	0.49	-0.11 to +0.03	0.55	-0.12 to -0.01 *
Garo Hills	0.55	0.57	-0.03 to +0.06	0.59	-0.06 to 0.00
Shivalik	0.77	0.85	+0.04 to +0.13 *	0.80	0.00 to +0.13 *

5 Discussion and limitations

What the evidence supports. For forest loss, a signed NDVI-decrease score matches or beats Prithvi and a smaller RGB model in every area. For lake water change, pretrained embeddings beat the standard water index but not plain reflectance. We found no setting in which an embedding clearly beat the best simple baseline: unsupervised scores never did, and a trained readout of Prithvi, which beat the unsupervised baseline in two forest areas, was matched by a readout of raw bands.

What it does not. Five forest areas, three chosen by a label-only search for active clearing, and two substantially changing lakes, chosen because radar shows change at them, are too few to settle the question; three scenes per period is a small noise sample; and urban or wetland change, which needs other reference products, was not tested. An unsupervised embedding score gives no direction of change, a real limit for loss detection that the probe sidesteps by seeing the sign. Every reference is another model's output, so all scores are agreement, not accuracy. Fine-tuning, the usual way to deploy such models, was not tested. A radar water model with conformal lake-area intervals, built on the same lakes, is described in the technical report [10].

6 Conclusion

Under a common noise standard, foundation-model embeddings did not outperform a signed vegetation baseline on forest loss, and their advantage over a water index at glacial lakes was matched by raw reflectance. A linear probe on Prithvi recovers an advantage over the unsupervised baseline in two areas, but a probe on six raw bands does about as well, so pretraining is not shown to help. The embeddings' low false-alarm rate in a placebo test is a calibration property, not evidence of detection skill. We release the pipeline, the evaluation design and all recorded outputs so that others can test further fields and fine-tuned models under the same standard.

Reproducibility. Code, scene lists, weight checksums and the exact numbers behind every table are in the project repository (`docs/V3_RESULTS.md`, `data/experiments/`), archived with a DOI [10]. Sentinel data are free and open (Copernicus); Hansen and Hi-MAG are CC-BY 4.0; Prithvi-EO-2.0 is Apache-2.0; the SSL4EO weights are CC-BY 4.0.

References

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